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Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

23CSE498 **Project Phase II** **2026**

B.Tech CSE – Initial Project Idea Submission

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1. Project Title

Runtime Computational Memory: Understanding Emergent Behaviour through KV-Cache Dynamics in Transformer Language Models

2. Project Abstract

2.1 Motivation

Large language models (LLMs) now mediate search, education, hiring, healthcare triage, and customer service, yet they continue to exhibit harmful emergent behaviours: social bias, hallucinations that persist and "snowball" across a conversation, and opinion reinforcement in multi-agent systems. Existing explanations attribute these behaviours almost entirely to the trained

parameters, and mitigation therefore relies on expensive retraining or fine-tuning. However, during inference a Transformer maintains a second, constantly evolving state, the Key–Value (KV) cache, which grows with every generated token and directly conditions all future attention. Systems research has already shown that this runtime state is highly structured: a small set of "heavy-hitter" entries dominates attention allocation, and initial-token "attention sinks" persist indefinitely. Yet no existing work asks *what* these persistent structures compute or *how* they shape downstream reasoning and behaviour. If long-horizon behaviours such as bias amplification and hallucination persistence are partly properties of runtime memory rather than weights alone, then they can be detected, attributed, and corrected at inference time, without retraining. The beneficiaries are broad: AI safety and alignment researchers gain a new mechanistic lens; model deployers gain cheap, targeted runtime interventions; and end users, especially members of groups affected by model bias, gain more reliable and fairer systems. This project is motivated by the opportunity to turn the KV cache from an engineering optimization into a first-class scientific object.

2.2 Problem Statement

Current Transformer research treats the KV cache purely as a decoding optimization: efficiency work compresses or evicts it, and interpretability work studies weights and activations while ignoring runtime state altogether. Consequently, there is no framework for (i) observing how runtime memory evolves over long, multi-turn and multi-agent conversations, (ii) attributing emergent behaviours such as social bias and persistent hallucination to identifiable structures within that memory, or (iii) causally intervening on runtime memory alone to change model behaviour. This project addresses that gap. Using social bias as the primary case study, it will instrument the KV cache of open-weight LLMs with full provenance (layer, head, token, turn, agent), test whether persistent *Memory Attractors* and localized *Bias Reservoirs* emerge during inference, whether bias propagates between interacting agents through shared runtime memory (*Memory Contagion*), whether bias accumulation exhibits phase-transition behaviour, and whether targeted KV-level interventions can alter reasoning without any parameter updates. The scope is restricted to open-weight models of 7–8B parameters and controlled multi-agent conversation settings, making the project feasible within the academic year.

3. Objectives

1. **Design and develop** a runtime-memory instrumentation framework that records per-token KV-cache state with full provenance (layer, head, attention allocation, turn, agent identity) during multi-agent LLM conversations.

2. **Detect and characterize** persistent **Memory Attractors** in long conversations using metrics such as KV reuse score, memory lifetime, attention persistence, and centrality.

3. **Analyze** whether social bias is localized into identifiable **Bias Reservoirs** — specific layers, heads, and KV regions that disproportionately accumulate and replay biased representations.

4. **Evaluate Memory Contagion** and phase-transition behaviour by comparing bias propagation across shared-memory versus independent-memory multi-agent settings under increasing semantic pressure.

5. **Demonstrate causal control** of model behaviour through KV-cache-only interventions (ablation, perturbation, gating), measuring changes in bias, hallucination, and reasoning coherence without any parameter modification.

4. Dataset / Data Source (if applicable)

The primary data — runtime memory traces — will be generated by the project itself by instrumenting open-weight LLMs (e.g., Llama-3-8B-Instruct, Mistral-7B-Instruct, Qwen2.5-7B; all publicly available on Hugging Face) during controlled multi-agent conversations. Conversation seeds and bias probes will be drawn from established public benchmarks: BBQ (`https://github.com/nyu-mll/BBQ`) and StereoSet (`https://github.com/moinnadeem/StereoSet`) for social bias, CrowS-Pairs for stereotype measurement, and TruthfulQA (`https://github.com/sylinrl/TruthfulQA`) for the hallucination case study. The resulting corpus of annotated KV-cache traces (with layer/head/turn/agent provenance) will itself be released as a public dataset deliverable. No proprietary data is required.

5. Expected Deliverables

1. An open-source runtime-memory instrumentation and multi-agent simulation framework (Python/PyTorch, Hugging Face Transformers) with a visualization dashboard for KV-cache dynamics.

2. A public annotated dataset of runtime memory traces from controlled multi-agent conversations, together with a causal KV-intervention toolkit (ablation, perturbation, gating).

3. A research paper reporting empirical evidence for (or against) *Memory Attractors*, *Bias Reservoirs*, *Memory Contagion*, and memory phase transitions, submitted to a peer-reviewed venue.

6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	Efficient Streaming Language Models with Attention Sinks (Xiao et al., ICLR)	2024	Shows that KV entries of initial tokens act as "attention sinks" that must be retained indefinitely for stable generation — direct evidence of persistent structures in runtime memory.
2	H2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models (Zhang et al., NeurIPS)	2023	Demonstrates that a small set of "heavy-hitter" KV entries dominates attention allocation and suffices to preserve generation quality under aggressive cache eviction.
3	Transformers are Multi-State RNNs (Oren et al., EMNLP)	2024	Reinterprets decoder-only Transformers as recurrent networks whose hidden state is the KV cache, formally framing the cache as the model's evolving state.
4	Function Vectors in Large Language Models (Todd et al., ICLR)	2024	Shows that compact, causally effective vector representations of tasks exist inside hidden states and can be transplanted to steer behaviour — evidence that inference-time state carries reusable computation.
5	Improving Factuality and Reasoning in Language Models through Multiagent Debate (Du et al., ICML)	2024	Shows that interaction between multiple LLM agents systematically shifts individual agents' answers, establishing that multi-agent dynamics reshape model behaviour.

6.1 Comparison of Related Work

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6.2 Novelty / Research Gap:

Across the compared papers, the KV cache is treated exclusively as an efficiency problem: attention sinks [1], heavy hitters [2], and the multi-state-RNN view [3] all establish that runtime memory is highly structured, yet none of them asks *what* those persistent structures compute or *how* they shape downstream behaviour. Interpretability work such as function vectors [4] locates causal structure in activations, but only within single forward passes, never in the temporally evolving KV state of a long conversation. Multi-agent studies [5] show that agent interaction changes outputs, but measure only surface text and offer no memory-level mechanism. No existing work instruments the KV cache with full provenance (layer, head, turn, agent), tests for persistent *Memory Attractors* or localized *Bias Reservoirs*, traces cross-agent *Memory Contagion*, or performs causal KV-only interventions. This project fills exactly that gap: it elevates runtime memory to a first-class research object and, using social bias as the case study, provides the first framework for observing, attributing, and causally editing emergent behaviour at the level of runtime memory — without any parameter updates. This directly motivates Objectives 1–5 in Section 3.

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Xiao et al. [1]	2024	StreamingLLM: retain attention-sink KV entries plus sliding window during decoding	PG-19, long-text streaming benchmarks	Sinks studied only for efficiency/stability; the semantic content and behavioural effect of persistent KV entries are never analysed
2	Zhang et al. [2]	2023	Dynamic KV eviction guided by cumulative-attention heavy-hitter scores	OpenBookQA, XSUM, and other LM benchmarks	Identifies dominant KV entries purely for compression; no link to bias, hallucination, or reasoning; single-agent, short-horizon settings
3	Oren et al. [3]	2024	Theoretical reframing of decoders as multi-state RNNs; TOVA cache-compression policy	Long-range language modelling benchmarks	Conceptual/ efficiency focus; does not track how the state evolves semantically over long dialogues or connect state to emergent behaviour
4	Todd et al. [4]	2024	Causal mediation to extract and transplant function vectors from attention heads	In-context learning task suites (word pairs, translation, etc.)	Operates on activations in single forward passes; does not study the temporal evolution of KV memory or multi-turn behavioural effects
5	Du et al. [5]	2024	Multi-round debate among LLM agents exchanging natural-language messages	GSM8K, MMLU, arithmetic and biography tasks	Measures only text-level outcomes (accuracy); provides no memory-level mechanism for how influence propagates, and ignores bias propagation

Table 1: Comparison of related work across key aspects

7. Potential Publication Venues with Justification:

Conferences: (1) **ACL / EMNLP** — premier NLP venues that regularly publish mechanistic interpretability and bias/fairness work on LLMs, matching our bias case study; (2) **ICLR** — the main venue for representation-level and inference-time analyses of Transformers (attention sinks and function vectors both appeared here); (3) **CODS-COMAD (India)** — ACM India's data science and AI conference, appropriate for the framework, dataset, and empirical-analysis contributions; **INDICON** is a further Indian fallback.

Journals: (1) **IEEE Transactions on Artificial Intelligence** — publishes foundational analyses of AI system behaviour and trustworthiness; (2) **Neurocomputing** — regularly publishes attention-mechanism and KV-cache analysis papers; (3) **IEEE Access** — broad-scope, suitable for the open-source instrumentation framework and reproducible dataset contribution.

8. References

[1] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient streaming language models with attention sinks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024. [Online]. Available: `https://arxiv.org/abs/2309.17453`

[2] Z. Zhang *et al.*, "H2O: Heavy-hitter oracle for efficient generative inference of large language models," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [Online]. Available: `https://arxiv.org/abs/2306.14048`

[3] M. Oren, M. Hassid, N. Yarden, Y. Adi, and R. Schwartz, "Transformers are multi-state RNNs," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2024. [Online]. Available: `https://arxiv.org/abs/2401.06104`

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[5] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. Int. Conf. Machine Learning (ICML)*, 2024. [Online]. Available: `https://arxiv.org/abs/2305.14325`

[6] I. O. Gallegos *et al.*, "Bias and fairness in large language models: A survey," *Computational Linguistics*, vol. 50, no. 3, pp. 1097–1179, 2024, doi: 10.1162/coli_a_00524.

[7] A. Parrish *et al.*, "BBQ: A hand-built bias benchmark for question answering," in *Findings of the Association for Computational Linguistics: ACL*, 2022, pp. 2086–2105. [Online]. Available: `https://aclanthology.org/2022.findings-acl.165`

[8] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in *Proc. 60th Annu. Meeting Assoc. Computational Linguistics (ACL)*, 2022, pp. 3214–3252. [Online]. Available: `https://aclanthology.org/2022.acl-long.229`

8. Faculty Remarks

As guide, I hereby confirm that I checked the plagiarism and AI content.

Note: The submitted document must have a similarity index of less than 15% and an AI-generated content score of less than 20%, as verified using Turnitin.

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